

Electrical Circuits

Lec 1: DC Circuits- Basic Concepts

Prof. Dr. Saeed Mahmud Ullah
EEE, DU

Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
- 5th edition

Electrical Circuits: Concepts

- An **electric circuit** is an interconnection of electrical elements.
- A simple electric circuit is shown in Fig. 1.1. It consists of three basic elements: a battery, a lamp, and connecting wires.

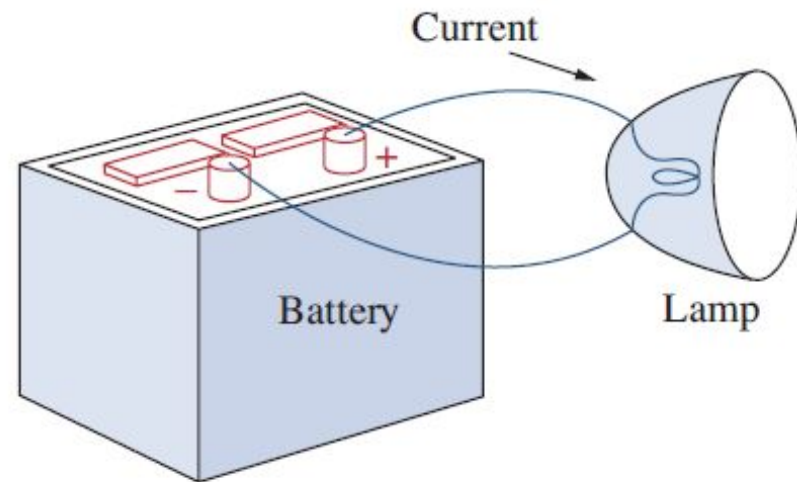


Figure 1.1
A simple electric circuit.

Circuit examples

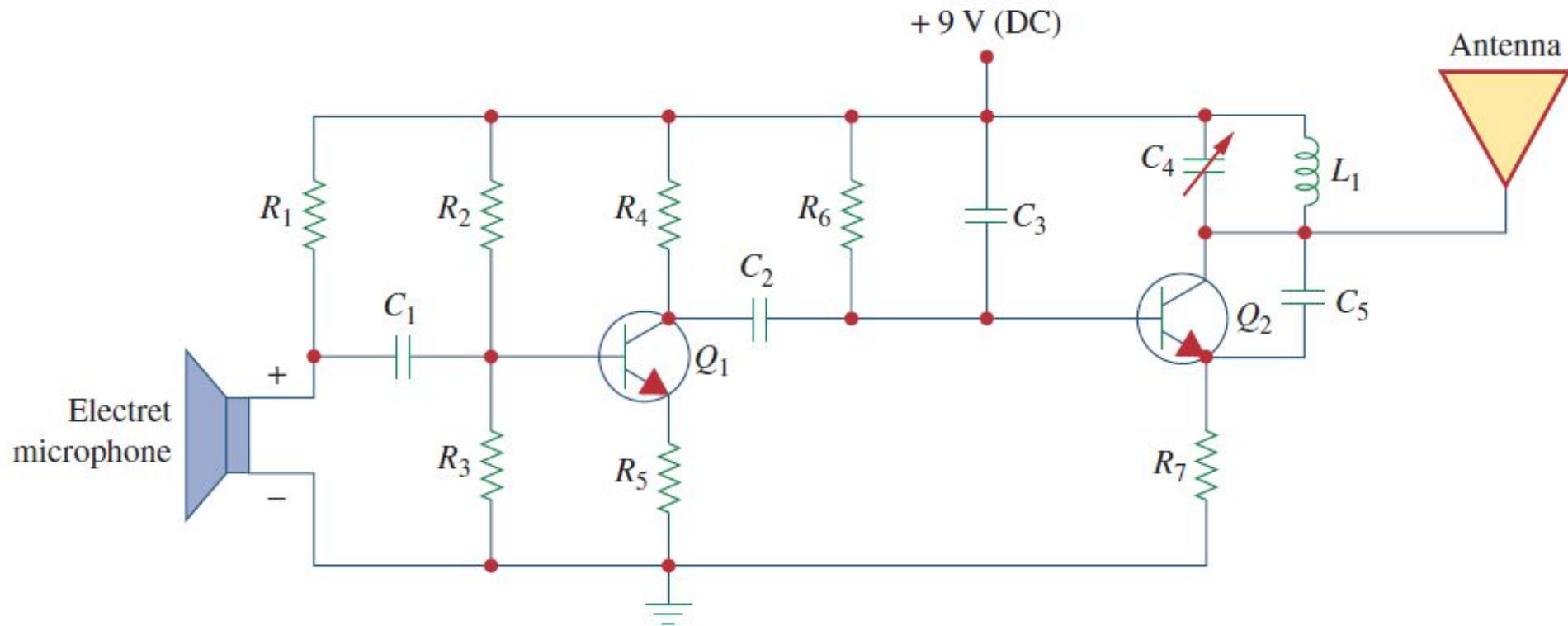


Figure 1.2

Electric circuit of a radio transmitter.

Charge and Current

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).
- We also know that the charge e on an electron is negative and equal in magnitude to 1.062×10^{-19} C, while a proton carries a positive charge of the same magnitude as the electron.
- The presence of equal numbers of protons and electrons leaves an atom neutrally charged.
- The following points should be noted about electric charge:
- The coulomb is a large unit for charges. In 1 C of charge, there are $1/(1.062 \times 10^{-19}) = 6.24 \times 10^{18}$ electrons.
- The *law of conservation of charge* states that charge can neither be created nor destroyed, only transferred.
- Thus the algebraic sum of the electric charges in a system does not change.

- **Electric current** is the time rate of change of charge, measured in amperes (A).
- Mathematically, the relationship between current i , charge q , and time t is

$$i \triangleq \frac{dq}{dt} \quad (1.1)$$

where current is measured in amperes (A), and 1 ampere = 1 coulomb/second

- The charge transferred between time and t is obtained by integrating both sides of Eq. (1.1). We obtain

$$Q \triangleq \int_{t_0}^t i \, dt \quad (1.2)$$

DC and AC

- A **direct current** (dc) is a current that remains constant with time.
- An **alternating current** (ac) is a current that varies sinusoidally with time.

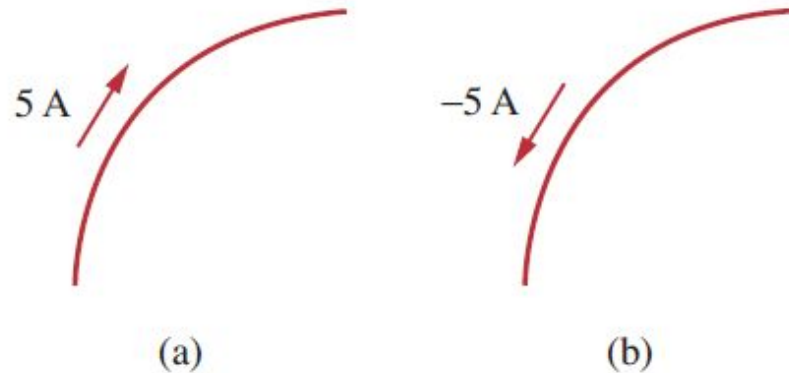
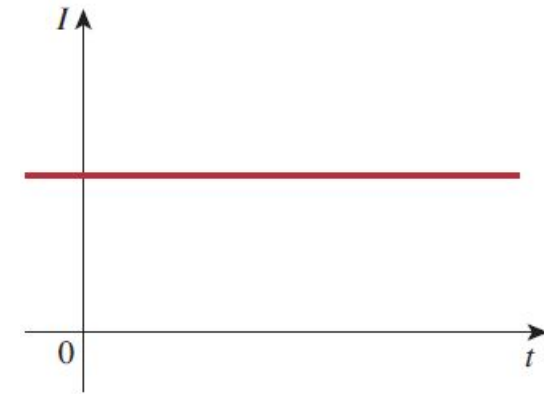
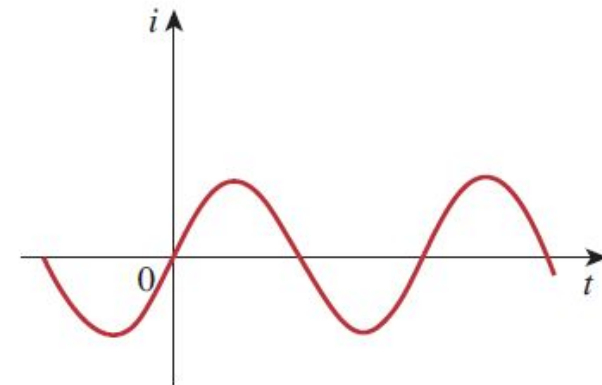


Figure 1.5
Conventional current flow: (a) positive current flow, (b) negative current flow.



(a)



(b)

Figure 1.4

Two common types of current: (a) direct current (dc), (b) alternating current (ac).

Example 1.1

How much charge is represented by 4,600 electrons?

Solution:

Each electron has -1.602×10^{-19} C. Hence 4,600 electrons will have

$$-1.602 \times 10^{-19} \text{ C/electron} \times 4,600 \text{ electrons} = -7.369 \times 10^{-16} \text{ C}$$

Practice Problem 1.1

Calculate the amount of charge represented by six million protons.

Answer: $+9.612 \times 10^{-13}$ C.

Example 1.2

The total charge entering a terminal is given by $q = 5t \sin 4\pi t$ mC. Calculate the current at $t = 0.5$ s.

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt}(5t \sin 4\pi t) \text{ mC/s} = (5 \sin 4\pi t + 20\pi t \cos 4\pi t) \text{ mA}$$

At $t = 0.5$,

$$i = 5 \sin 2\pi + 10\pi \cos 2\pi = 0 + 10\pi = 31.42 \text{ mA}$$

Practice Problem 1.2

If in Example 1.2, $q = (10 - 10e^{-2t})$ mC, find the current at $t = 1.0$ s.

Answer: 2.707 mA.

Example 1.3

Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is $i = (3t^2 - t)$ A.

Solution:

$$\begin{aligned} Q &= \int_{t=1}^2 i \, dt = \int_1^2 (3t^2 - t) \, dt \\ &= \left(t^3 - \frac{t^2}{2} \right) \Big|_1^2 = (8 - 2) - \left(1 - \frac{1}{2} \right) = 5.5 \text{ C} \end{aligned}$$

Practice Problem 1.3

The current flowing through an element is

$$i = \begin{cases} 4 \text{ A}, & 0 < t < 1 \\ 4t^2 \text{ A}, & t > 1 \end{cases}$$

Calculate the charge entering the element from $t = 0$ to $t = 2$ s.

Answer: 13.333 C.

Voltage

- To move an electron in a conductor in a particular direction requires some work or energy transfer.
- This work is performed by an external electromotive force (emf), typically represented by the battery in Fig. 1.3. This emf is also known as ***voltage or potential difference***.
- The voltage between two points a and b in an electric circuit is the energy (or work) needed to move a unit charge from a to b ; mathematically,

$$v_{ab} \triangleq \frac{dw}{dq} \quad (1.3)$$

- **Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).**
- Figure 1.6 shows the voltage across an element (represented by a rectangular block) connected to points a and b . The plus and minus signs are used to define reference direction or voltage polarity. The v_{ab} can be interpreted in two ways: (1) Point a is at a potential of v_{ab} volts higher than point b , or (2) the potential at point a with respect to point b is v_{ab} .

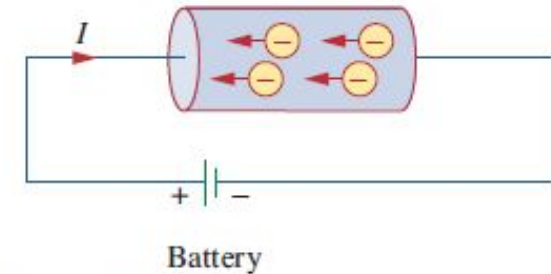


Figure 1.3
Electric current due to flow of electronic charge in a conductor.

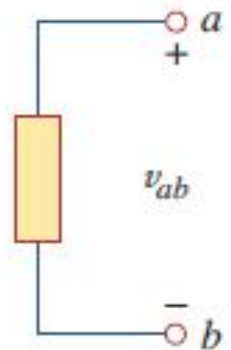


Figure 1.6
Polarity of voltage v_{ab} .

Power and Energy

- **Power** is the time rate of expending or absorbing energy, measured in watts (W).

$$p \triangleq \frac{dw}{dt} \quad (1.5)$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi \quad (1.6)$$

- The *law of conservation of energy* must be obeyed in any electric circuit. For this reason, the algebraic sum of power in a circuit, at any instant of time, must be zero:

$$\boxed{\sum p = 0} \quad (1.8)$$

- **Energy** is the capacity to do work, measured in joules (J).
- The energy absorbed or supplied by an element from time t_0 to time t is

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt \quad (1.9)$$

Example 1.4

An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

Solution:

The total charge is

$$\Delta q = i \Delta t = 2 \times 10 = 20 \text{ C}$$

The voltage drop is

$$v = \frac{\Delta w}{\Delta q} = \frac{2.3 \times 10^3}{20} = 115 \text{ V}$$

Example 1.5

Find the power delivered to an element at $t = 3$ ms if the current entering its positive terminal is

$$i = 5 \cos 60\pi t \text{ A}$$

and the voltage is: (a) $v = 3i$, (b) $v = 3 di/dt$.

Solution:

(a) The voltage is $v = 3i = 15 \cos 60\pi t$; hence, the power is

$$p = vi = 75 \cos^2 60\pi t \text{ W}$$

At $t = 3$ ms,

$$p = 75 \cos^2 (60\pi \times 3 \times 10^{-3}) = 75 \cos^2 0.18\pi = 53.48 \text{ W}$$

(b) We find the voltage and the power as

$$v = 3 \frac{di}{dt} = 3(-60\pi)5 \sin 60\pi t = -900\pi \sin 60\pi t \text{ V}$$

$$p = vi = -4500\pi \sin 60\pi t \cos 60\pi t \text{ W}$$

At $t = 3$ ms,

$$\begin{aligned} p &= -4500\pi \sin 0.18\pi \cos 0.18\pi \text{ W} \\ &= -14137.167 \sin 32.4^\circ \cos 32.4^\circ = -6.396 \text{ kW} \end{aligned}$$

Example 1.6

How much energy does a 100-W electric bulb consume in two hours?

Solution:

$$\begin{aligned}w &= pt = 100 \text{ (W)} \times 2 \text{ (h)} \times 60 \text{ (min/h)} \times 60 \text{ (s/min)} \\&= 720,000 \text{ J} = 720 \text{ kJ}\end{aligned}$$

This is the same as

$$w = pt = 100 \text{ W} \times 2 \text{ h} = 200 \text{ Wh}$$

Circuit Elements

- There are two types of elements found in electric circuits: ***passive* elements** and ***active* elements**.
- An active element is capable of generating energy while a passive element is not.
- Examples of passive elements are resistors, capacitors, and inductors.
- Typical active elements include generators, batteries, and operational amplifiers.
- An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

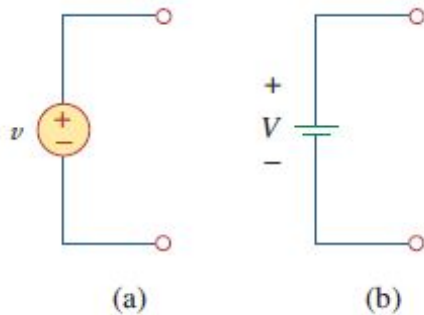


Figure 1.11

Symbols for independent voltage sources:
(a) used for constant or time-varying voltage, (b) used for constant voltage (dc).

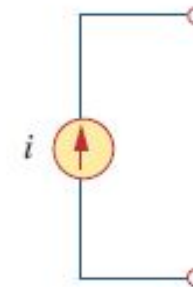


Figure 1.12

Symbol for independent current source.

- An **ideal dependent** (or **controlled**) **source** is an active element in which the source quantity is controlled by another voltage or current.

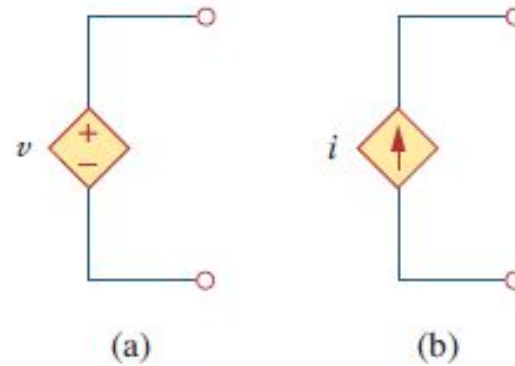


Figure 1.13

Symbols for: (a) dependent voltage source, (b) dependent current source.

Example 1.7

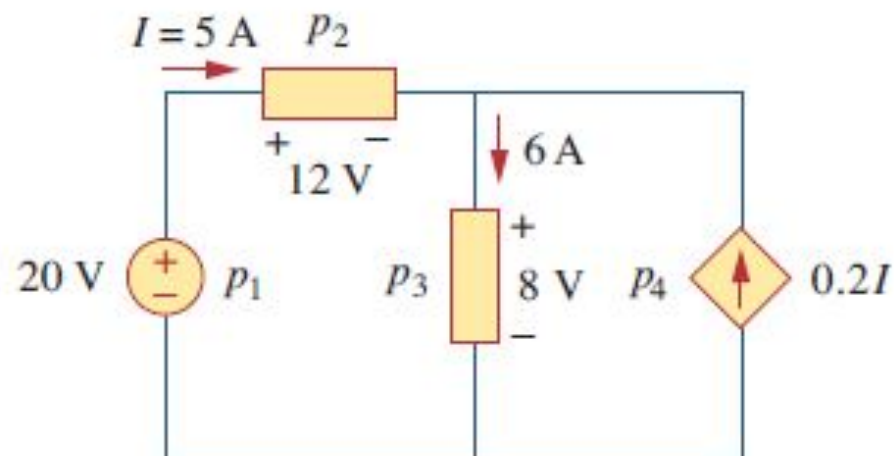


Figure 1.15
For Example 1.7.

Calculate the power supplied or absorbed by each element in Fig. 1.15.

Solution:

We apply the sign convention for power shown in Figs. 1.8 and 1.9. For p_1 , the 5-A current is out of the positive terminal (or into the negative terminal); hence,

$$p_1 = 20(-5) = -100 \text{ W} \quad \text{Supplied power}$$

For p_2 and p_3 , the current flows into the positive terminal of the element in each case.

$$p_2 = 12(5) = 60 \text{ W} \quad \text{Absorbed power}$$

$$p_3 = 8(6) = 48 \text{ W} \quad \text{Absorbed power}$$

For p_4 , we should note that the voltage is 8 V (positive at the top), the same as the voltage for p_3 , since both the passive element and the dependent source are connected to the same terminals. (Remember that voltage is always measured across an element in a circuit.) Since the current flows out of the positive terminal,

$$p_4 = 8(-0.2I) = 8(-0.2 \times 5) = -8 \text{ W} \quad \text{Supplied power}$$

We should observe that the 20-V independent voltage source and $0.2I$ dependent current source are supplying power to the rest of the network, while the two passive elements are absorbing power. Also,

$$p_1 + p_2 + p_3 + p_4 = -100 + 60 + 48 - 8 = 0$$

In agreement with Eq. (1.8), the total power supplied equals the total power absorbed.

Practice Problem 1.7

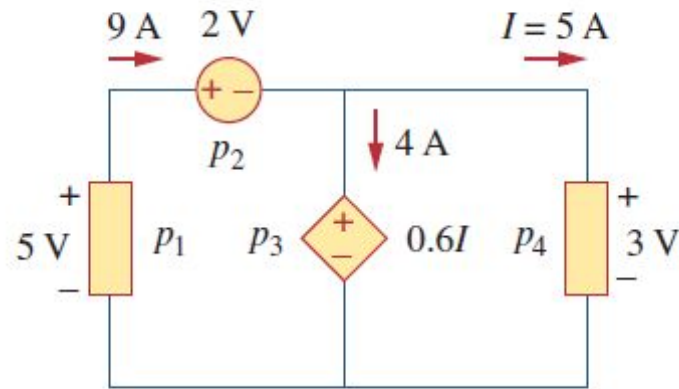


Figure 1.16

For Practice Prob. 1.7.

Compute the power absorbed or supplied by each component of the circuit in Fig. 1.16.

Answer: $p_1 = -45\text{ W}$, $p_2 = 18\text{ W}$, $p_3 = 12\text{ W}$, $p_4 = 15\text{ W}$.

1.18 Find the power absorbed by each of the elements in Fig. 1.29.

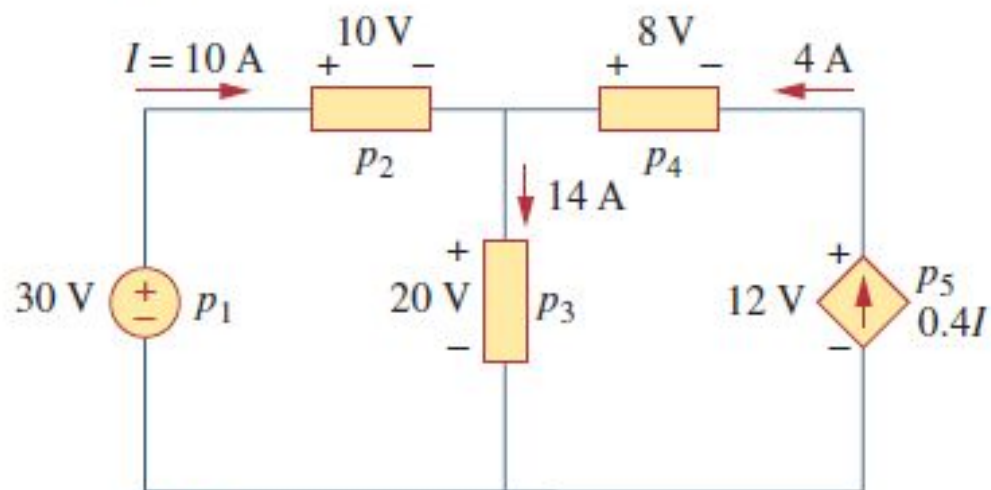


Figure 1.29
For Prob. 1.18.

1.20 Find V_o and the power absorbed by each element in the circuit of Fig. 1.31.

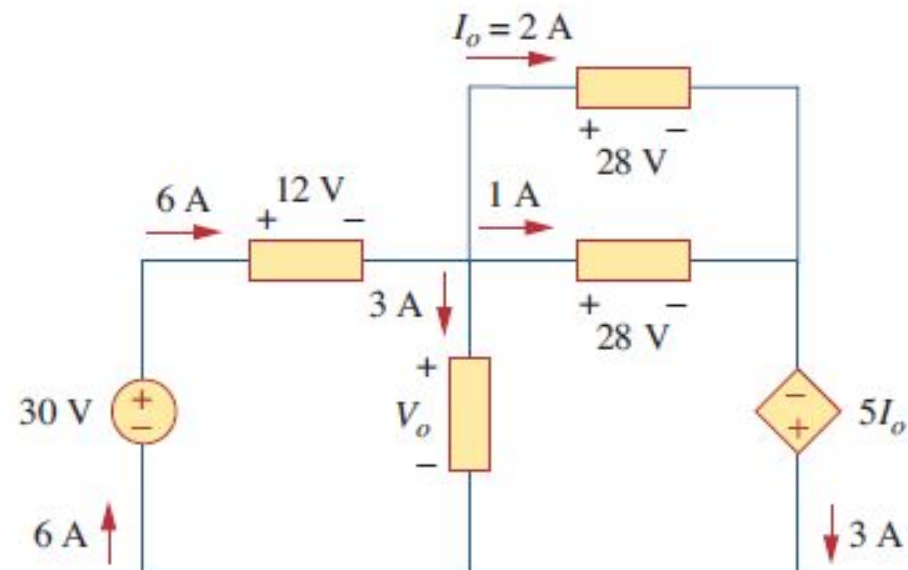


Figure 1.31
For Prob. 1.20.